

General form of sol'n of state eq'n:

$$\underline{x}(t) = \Phi(t, t_0) \underline{x}_0 + \int_{t_0}^t \Phi(t, \tau) B(\tau) \underline{u}(\tau) d\tau$$

where Φ is the state transition matrix (STM)

with $\Phi(t, \tau) = \Psi(t) \Psi(\tau)^{-1}$, Ψ fundamental

= Properties of any STM:

1.) $\Phi(t, t) = \Phi(\tau, \tau) = I$ for any t, τ .

2.) $\Phi(t, \tau)$ nonsingular for any t, τ

$$\Rightarrow \det \Phi(t, \tau) \neq 0$$

$$\begin{aligned} \det \Phi(t, \tau) &= \det [\Psi(t) \Psi(\tau)^{-1}] \\ &= \det \Psi(t) \det \Psi(\tau)^{-1} \\ &= \det \Psi(t) [\det \Psi(\tau)]^{-1} \end{aligned}$$

But we know $\det \Psi(t) \neq 0$

and same proof shows:

$$\det \Psi(t) \neq \infty \text{ at any finite } t$$

$$\Rightarrow \det \Phi(t, \tau) \neq 0 \checkmark$$

$$3.) \Phi(t, \tau)^{-1} = \Phi(\tau, t)$$

$$\begin{aligned} \text{Pf: } \Phi(t, \tau)^{-1} &= [\Psi(t) \Psi(\tau)^{-1}]^{-1} \\ &= (\Psi(\tau)^{-1})^{-1} \Psi^{-1}(t) \\ &= \Psi(\tau) \Psi^{-1}(t) \\ &= \Phi(\tau, t) \checkmark \end{aligned}$$

4.)

a.) $\frac{\partial}{\partial t} \Phi(t, \tau) = A(t) \Phi(t, \tau)$

b.) $\frac{\partial}{\partial \tau} \Phi(t, \tau) = -\Phi(t, \tau) A(\tau)$

$$\begin{aligned} \text{Pf a.) } \frac{\partial}{\partial t} \Phi(t, \tau) &= \frac{\partial}{\partial t} [\Psi(t) \Psi(\tau)^{-1}] \\ &= \left[\frac{\partial}{\partial t} \Psi(t) \right] \Psi^{-1}(\tau) \\ &= A(t) \Psi(t) \Psi^{-1}(\tau) \\ &= A(t) \Phi(t, \tau) \checkmark \end{aligned}$$

$$\begin{aligned} \text{b.) } \frac{\partial}{\partial \tau} \Phi(t, \tau) &= \Psi(t) \left[\frac{\partial}{\partial \tau} \Psi^{-1}(\tau) \right] \\ &= \Psi(t) [-\Psi^{-1}(\tau) A(\tau)] \\ &= -\Phi(t, \tau) A(\tau) \checkmark \end{aligned}$$

5.) For any t_0, t_1, t_2

$$\Phi(t_2, t_0) = \Phi(t_2, t_1) \Phi(t_1, t_0)$$

"Composition" or "semi-group" property

pf $\Phi(t_2, t_0) = \Psi(t_2) \Psi(t_0)^{-1}$

$$\Phi(t_1, t_0) = \Psi(t_1) \Psi(t_0)^{-1}$$

$$\Phi(t_2, t_1) = \Psi(t_2) \Psi(t_1)^{-1}$$

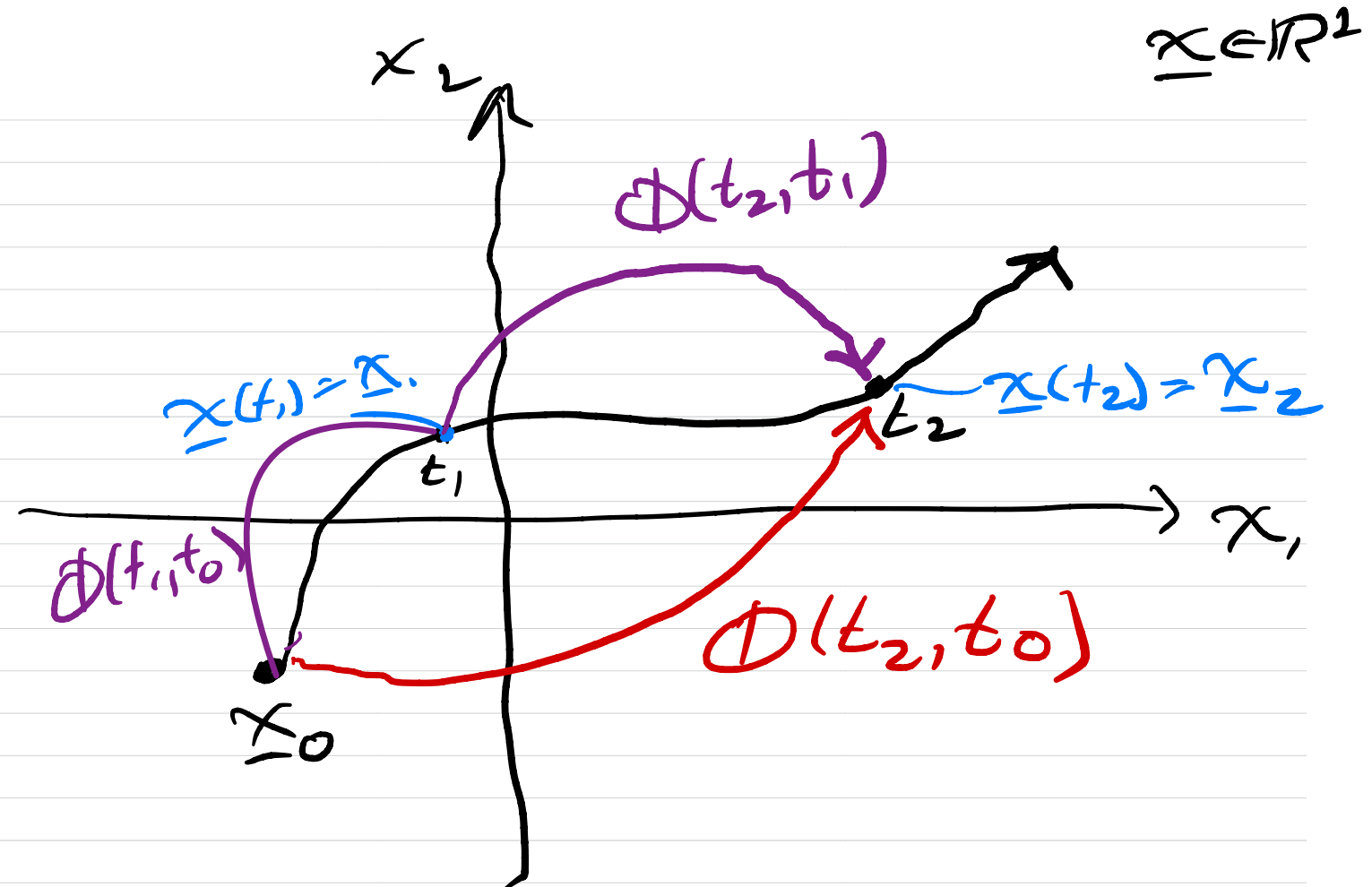
$$\Phi(t_2, t_1) \Phi(t_1, t_0) = [\Psi(t_2) \cancel{\Psi(t_1)^{-1}}] [\cancel{\Psi(t_1)} \Psi(t_0)^{-1}]$$

$$= \Psi(t_2) \Psi(t_0)^{-1}$$

$$= \Phi(t_2, t_0) \quad \checkmark$$

Intuitively, consider homop. sol'n

$$\underline{x}(t) = \Phi(t, t_0) \underline{x}_0 = \underline{x}(t_0)$$



$$\underline{x}_1 = \underline{x}(t_1) = \Phi(t_1, t_0) \underline{x}_0$$

$$\begin{aligned} \underline{x}_2 = \underline{x}(t_2) &= \Phi(t_2, t_0) \underline{x}_0 \\ &= \Phi(t_2, t_1) \underline{x}_1 \end{aligned}$$

Note also this implies

$$\begin{aligned} \Phi(t, \tau) &= \Phi(t, 0) \Phi(0, \tau) \\ &= \Phi(t, 0) \Phi(\tau, 0)^{-1} \end{aligned}$$

Can we actually constructively determine the STM for an arbitrary $A(t)$?

i.e. find the n^2 f'ns of t, τ

Generally no, but can try:

1.) Find family of fund'l sol'ns

$\Rightarrow$ remember: $\dot{\underline{x}} = A \underline{x}$

$$\Leftrightarrow \begin{aligned} \dot{x}_1 &= a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \\ &\vdots \\ \dot{x}_n &= a_{n1}x_1 + \dots + a_{nn}x_n \end{aligned}$$

2.) Numerically solve $\Phi(t, t_0)$ on a grid of $t > t_0$ points.

3.) Brute force series expansion (may possibly simplify)

"Peano-Baker" series

$$\begin{aligned} \Phi(t, \tau) &= I + \int_{\tau}^t A(\sigma) d\sigma + \int_{\tau}^t A(\sigma_1) \int_{\tau}^{\sigma_1} A(\sigma_2) d\sigma_2 d\sigma_1 \\ &+ \int_{\tau}^t A(\sigma_1) \int_{\tau}^{\sigma_1} A(\sigma_2) \int_{\tau}^{\sigma_2} A(\sigma_3) d\sigma_3 d\sigma_2 d\sigma_1 \\ &+ \dots \text{(infinite \# terms)} \end{aligned}$$

IF $A(t) \int_{\tau}^t A(\sigma) d\sigma = \left[\int_{\tau}^t A(\sigma) d\sigma \right] A(t)$

(i.e. $A(t)$ commutes w/ its integral) for every t, τ

THEN

$$\Phi(t, \tau) = \sum_{k=0}^{\infty} \frac{1}{k!} M(t, \tau)^k$$

where

$$M(t, \tau) = \int_{\tau}^t A(\sigma) d\sigma$$

= recall scalar exponential f'n satisfies

$$e^m = \sum_{k=0}^{\infty} \frac{m^k}{k!}$$

By analogy the above power series defines matrix exp. f'n

Common to write (when commutativity
cond'n is satisfied!)

$$\Phi(t, \tau) = \exp \left[\int_{\tau}^t A(\sigma) d\sigma \right]$$

where by "exp" is meant the
formal power series above.

— This looks comforting familiar,
But must be careful.

= for example, if

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$e^M \neq \begin{bmatrix} e^a & e^b \\ e^c & e^d \end{bmatrix} \quad \text{No!}$$

So what does e^M look like?
(later)

As a final possible simplification,
if A is constant, then the
commutativity cond'n is definitely
satisfied, and

$$\begin{aligned} M(t, \tau) &= \int_{\tau}^t A d\sigma \\ &= A \int_{\tau}^t d\sigma \end{aligned}$$

$$M(t, \tau) = (t - \tau) A$$

$\Rightarrow$

$$\Phi(t, \tau) = \exp[(t - \tau) A]$$

$$= e^{(t - \tau) A}$$

(Note: can also write
 $e^{A(t - \tau)}$ when A understood
to be constant.)

Properties of matrix exp, constant A:

$$e^{At} = \Phi(t, 0) \text{ when } A \text{ is const}$$

- $e^{At}|_{t=0} = \Phi(0, 0) = \mathbb{I}$
- e^{At} nonsingular for any finite t
- $e^{A(t_1+t_2)} = e^{At_1} e^{At_2}$
- $[e^{At}]^{-1} = e^{-At}$
- $[e^{At}]^T = e^{A^T t}$
- $e^{(A+B)t} \neq e^{At} e^{Bt}$

unless A, B commute
 $AB = BA$

Now, consider:

$$A_1 = \begin{bmatrix} -1 & 2 \\ -2 & -1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 0 & 1 \\ -5 & -2 \end{bmatrix}$$

$e^{A_1 t}$ and $e^{A_2 t}$ will both have entries that are combinations of the scalar f'ns

$$e^{-t} \cos 2t, e^{-t} \sin 2t$$

How to generally deduce facts like this, and to systematically construct e^{At} for a given A?

$\Rightarrow$ Next Time